2303A52416
batch=35
Task 1 – Customer Data Cleaning
Task:
Use AI to generate a Python script for cleaning a customer dataset.
Instructions:
• Handle missing values in columns (age, income, city).
• Remove duplicate records.
• Standardize text values (e.g., city names in lowercase).
• Encode categorical variables (gender, city).
Sample Code:

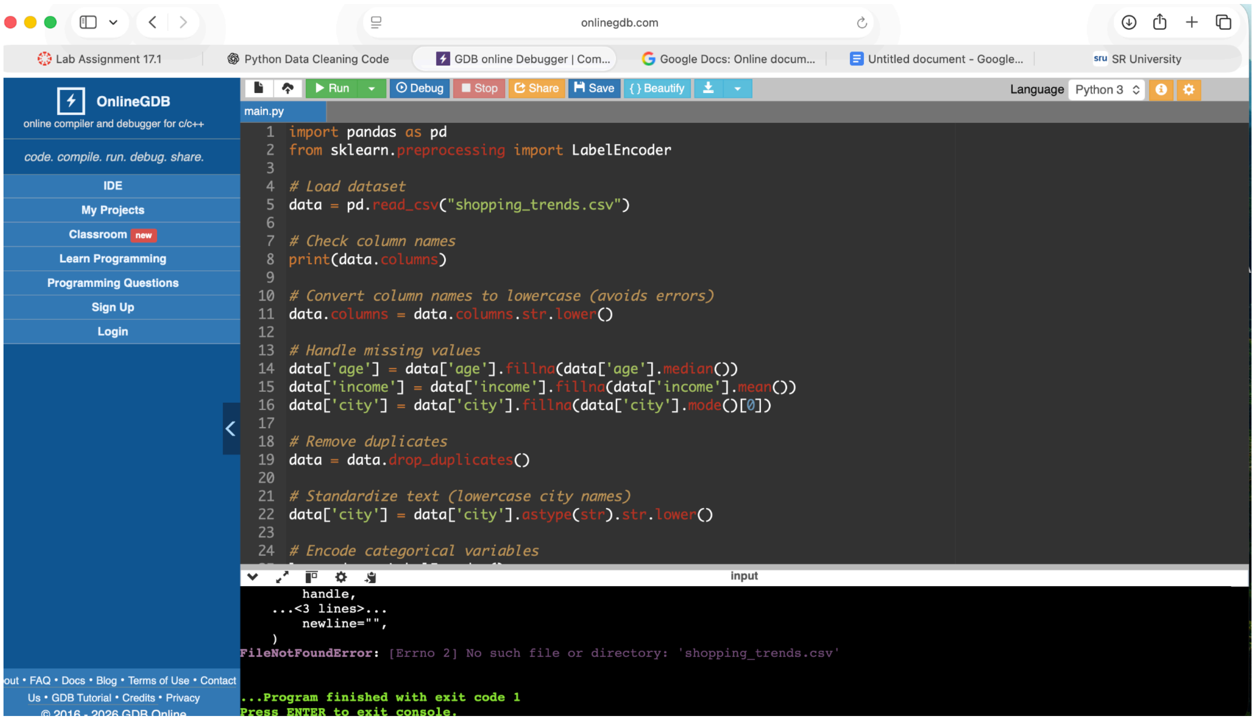
```
import pandas as pd
data = pd.read_csv("shopping_trends.csv")
print(data.head())
```

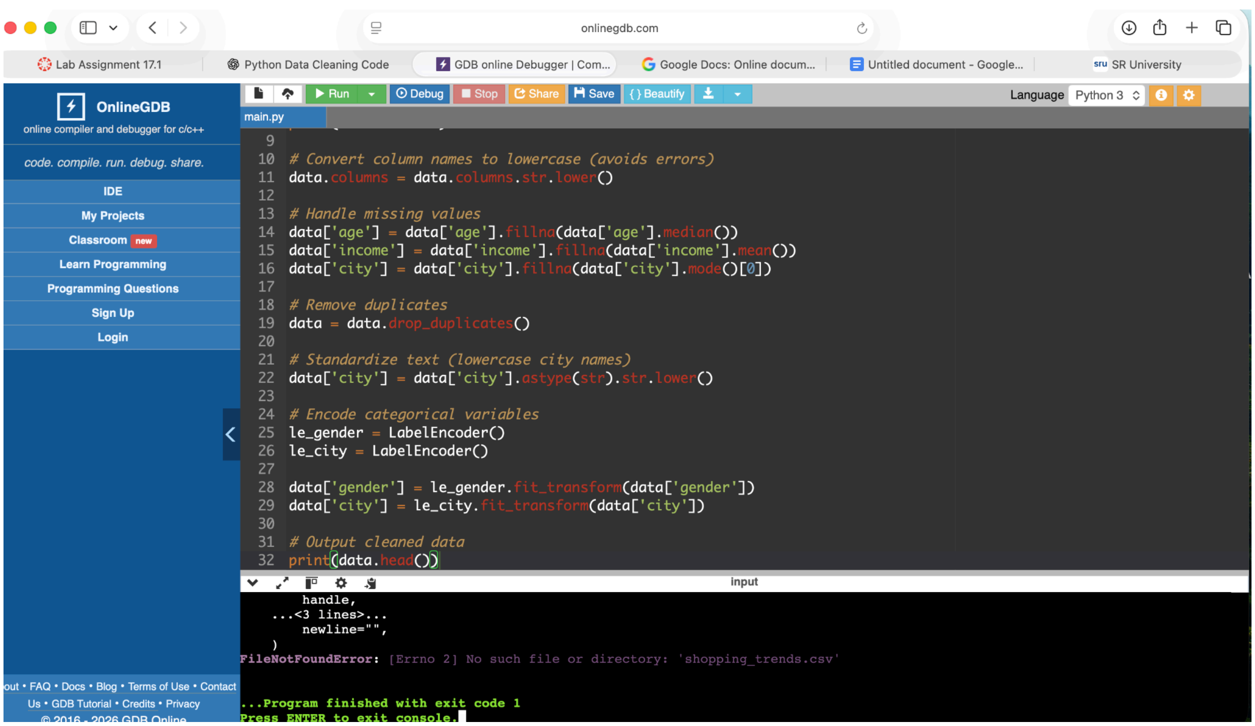
Dataset link:
https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset?select=shopping_trends.csv
Expected Output:
• A cleaned Pandas data frame ready for analysis.
Code

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Load dataset
data = pd.read_csv("shopping_trends.csv")

# Handle missing values
data['age'].fillna(data['age'].median(), inplace=True)
data['income'].fillna(data['income'].mean(), inplace=True)
data['city'].fillna(data['city'].mode()[0], inplace=True)

# Remove duplicates
data.drop_duplicates(inplace=True)

# Standardize text (lowercase city names)
data['city'] = data['city'].str.lower()

# Encode categorical variables
le = LabelEncoder()
```

data['gender'] = le.fit_transform(data['gender'])
data['city'] = le.fit_transform(data['city'])

print(data.head())





Task 2 – Sales Data Preprocessing

Task:

Preprocess sales transaction data using AI assistance.
Instructions:

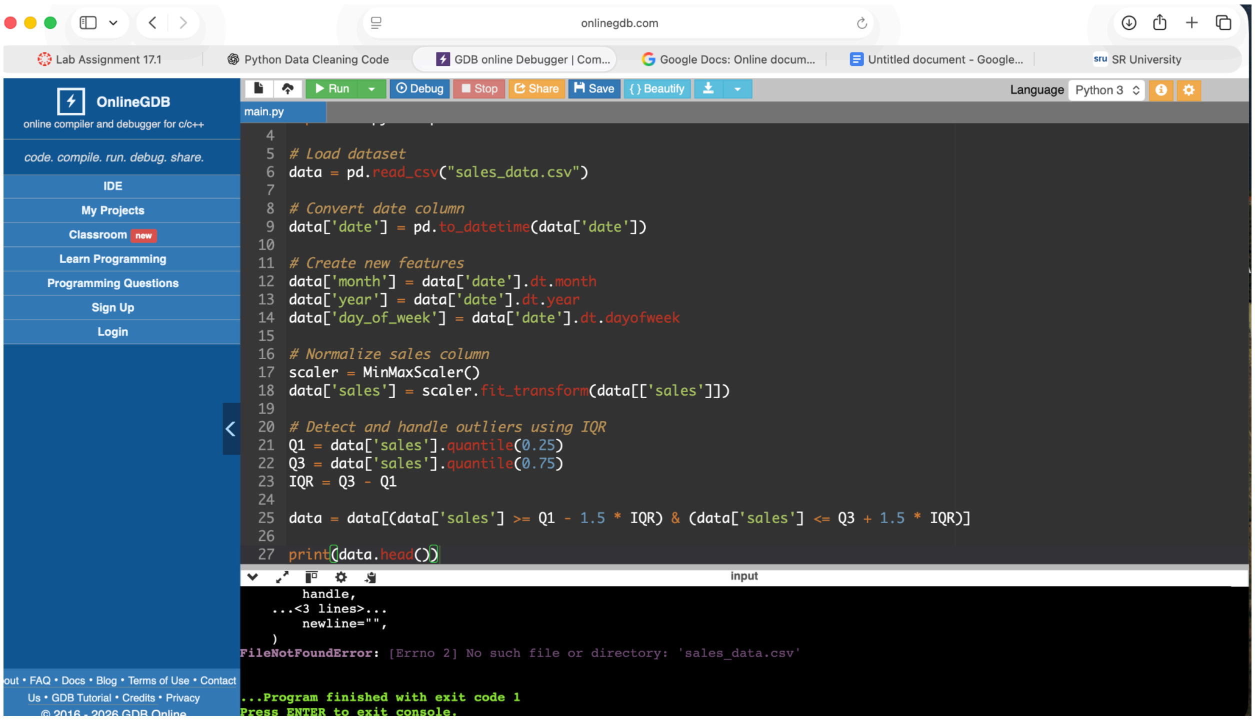
- Convert date columns into datetime format.
- Create new features (month, year, day of week).
- Normalize sales amount column.
- Detect and handle outliers in dataset.

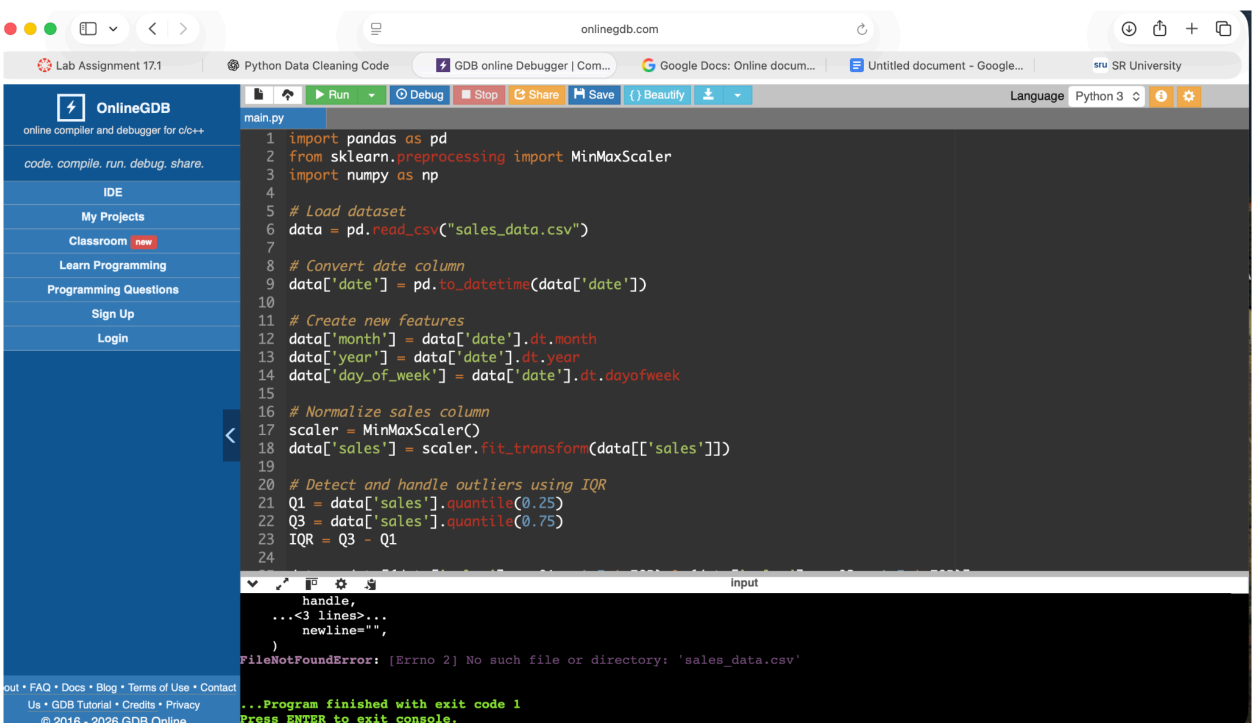
Sample Code:

```
import pandas as pd
data = pd.read_csv("sales_data.csv")
print(data.head())
```

Dataset link:
https://www.kaggle.com/datasets/vinothkannaece/sales-dataset?select=sales_data.csv
Expected Output:

- A transformed dataset with time-based features and normalized values.





Task 3 – Healthcare Data Preparation

Task:

Clean and preprocess a healthcare dataset using AI scripts.

Instructions:

• Handle missing values in blood_pressure, cholesterol.

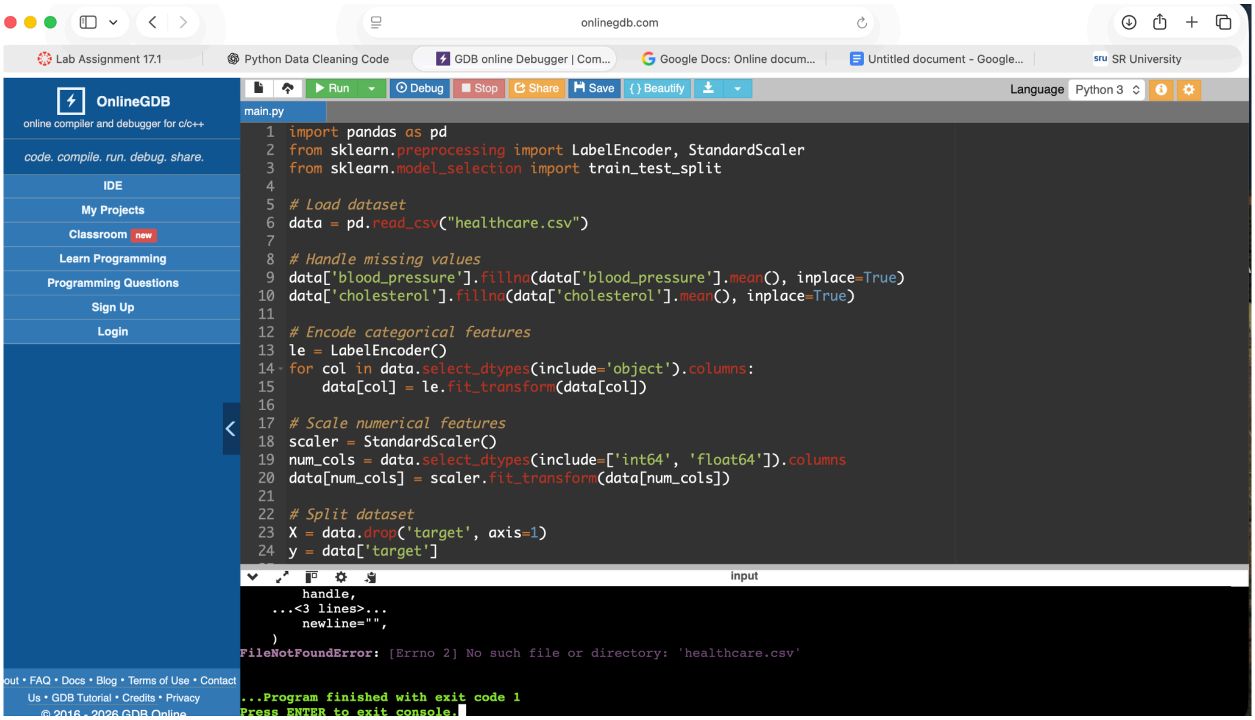
• Encode categorical features
• Scale numerical features (min-max or standard scaling).
• Split dataset into training and testing sets.

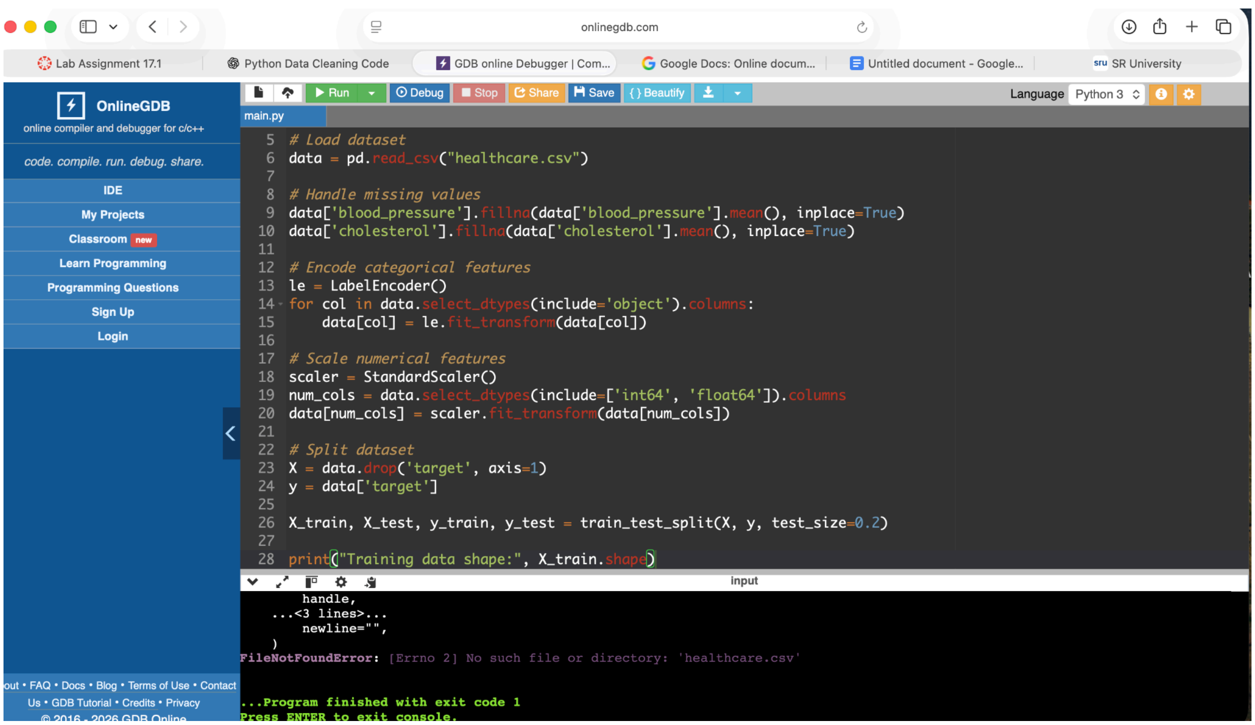
Dataset link :
https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset

Expected Output:
• A preprocessed dataset ready for machine learning models.

Code

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split

# Load dataset
data = pd.read_csv("healthcare.csv")

# Handle missing values
data['blood_pressure'].fillna(data['blood_pressure'].mean(), inplace=True)
data['cholesterol'].fillna(data['cholesterol'].mean(), inplace=True)

# Encode categorical features
le = LabelEncoder()
for col in data.select_dtypes(include='object').columns:
    data[col] = le.fit_transform(data[col])

# Scale numerical features
scaler = StandardScaler()
num_cols = data.select_dtypes(include=['int64', 'float64']).columns
data[num_cols] = scaler.fit_transform(data[num_cols])

# Split dataset
X = data.drop('target', axis=1)
y = data['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

print("Training data shape:", X_train.shape)
```





Task 4 – Employee Salary Data Preprocessing

Task:

Clean and preprocess an employee salary dataset using AI scripts.

Instructions:

- Handle missing values
- Detect and remove salary outliers.

• Standardize department names (lowercase).
• Apply label encoding to job location, department.
Dataset link:
https://www.kaggle.com/code/vishantmathur/employee-salary
Expected Output:
• A structured dataset ready for HR analytics.

Code